Astrophysical Lab: Winter Semester 2026 / 2027

Title of experiment: Adaptive Optics

Supervisor (1st afternoon): Frank Grupp

Supervisor (2nd afternoon): Frank Grupp

Group (number): 6

Names (of the students): Max Ammons, Catherine Lambe, Wang Zihan, Craig Arnott

(Please also indicate whether you are a Master's student in Astrophysics (MA) or in Physics (MP) and in which semester you are.)

1st afternoon (date, times of start and end): 20/10/26 13:15-18:30

2nd afternoon (date, times of start and end): 27/10/26 13:15-18:30

Time you needed to work on the report (at home)
in person-hours, sum over all group members: 60

(This information does not enter into the evaluation, but it helps us to adjust the lab content.)

Here is space for your comments:

What did you like? How would you improve the labs? Would you actually like to implement these improvements yourself?

Signatures of the students, with date: Zihan Wang 06/11/26

Catherine Lambe.

With your signatures you confirm that you have not used any references for your work other than those quoted in your lab report, and that you have not copied any content from other students' reports.

Evaluation of the lab report and the performance of the students during the lab (by the supervisor):

Signature of the lab supervisor, with date:

Ludwig-Maximilians-Universität München

17141 Observational and Instrumental Lab

Experiment 1: Adaptive Optics

Group 6

By: Max Ammons, Craig Arnott, Catherine Lambe, Zihan Wang.

Abstract

The aim of this experiment was to investigate the ability of an adaptive optics system to correct for both static and turbulence-induced wavefront errors, and thus improve image quality. A key parameter used to determine image quality was the Strehl ratio ($0 \leq S \leq 1$), where the image quality of the system improves with S. While manually correcting the static WFEs increased the Strehl ratio from 0.0 ± 0.0 to 0.5683 ± 0.0002, applying the AO system's automatic-correction training loop resulted in $S = 0.9890603 \pm 0.0000004$. This improvement was seen in the corresponding science-camera image, which closely approximated an ideal Airy pattern. Potential astigmatism within the optical system was, however, indicated by the Airy pattern's noted elongation along the y-axis, with the Full-Width-Half-Maximum being 35.0 ± 1.0 pixels along x and 50.0 ± 1.0 pixels along y. The system's stability was investigated via the fluctuation of the Zernike coefficients as a function of time. For the automatically-corrected system, all coefficients from order Z0 to Z20 were recorded as being 0.00 μm at all time steps, but the limited instrument precision of 0.01 μm means that this cannot be taken as a guarantee of perfect system stability. Zernike coefficients were also varied to introduce controlled aberrations, allowing their visual effects to be analysed. The addition of a turbulence simulator significantly increased the overall WFE, yet automatic-correction by the AO system reduced both RMS and P2V WFE by approximately 94%. Once more, aberrations were not fully eliminated, with remaining WFEs of 0.213 μm in RMS and 0.835 μm in P2V.

1 Motivation

A telescope's angular resolution determines how well it can distinguish closely separated sources. Although larger apertures allow for finer angular resolution, for ground-based telescopes atmospheric turbulence and optical aberrations can limit this. Adaptive optics aims to reduce these effects and create an image with quality closer to the diffraction-limited case[1].

Temperature variations in the atmosphere produce variations in refractive index, thus distorting incoming, initially planar, wavefronts [2]. As these variations evolve, the distortions change. Long exposures average their effects, producing a broadened point-spread-function (PSF).

Adaptive optics is a rapid closed loop system, in which a wavefront sensor (WFS) measures the wavefront error (WFE), a controller calculates the required correction, and the deformable mirror (DM) is deformed via actuators to impose an opposite distortion[1]. The loop repeats, and progressively reduces the WFE and concentrates the light into a narrower higher-peak PSF.

In this experiment we aim to investigate wavefront measurement and correction, check system stability, and examine how deliberately introduced aberrations and simulated turbulence affects wavefronts and PSF.

2 Experimental Set-Up

The experimental set-up used in this experiment is shown in Fig. 1. The primary components included a 635nm laser source, a deformable mirror, a wavefront sensor, and a science camera CCD with a detection efficiency of 80% of the system wavelength[1].

The propagation of the produced rays through the adaptive optics system is illustrated in Fig. 2. The adaptive optics system was controlled using the desktop programme ALPAO on the connected computer, with the connection points illustrated in Fig. 2.

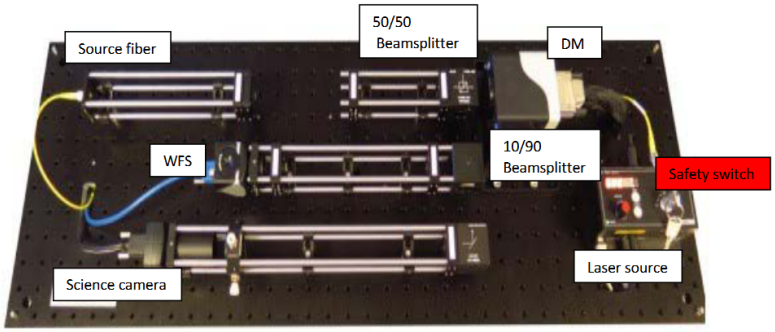


Figure 1: Photograph of the experimental apparatus used, which included a 635 nm laser source, a deformable mirror, a wavefront sensor, a science camera, and 50/50 and 10/90 beamsplitters[1]. A corresponding ray tracing diagram is illustrated in Fig. 2. The adaptive optics apparatus was primarily controlled using the desktop programme ALPAO on the connected computer[1].

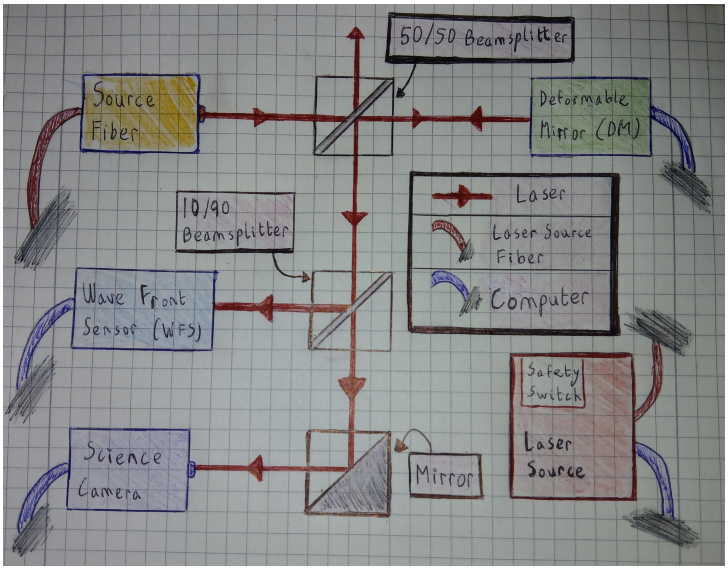


Figure 2: Illustration of the propagation of rays through the adaptive optics system used in this experiment and its components.

Assuming lossless transmission through all optical surfaces, perfect mirror reflectivity and a detection efficiency of 80% for the CCD, the efficiency, η_{ideal}, of the system was 18%.

$$\eta_{\text{ideal}} = 0.50 \times 0.50 \times 0.90 \times 0.80 = 0.18. \tag{1}$$

While operating the apparatus, laser safety was carefully maintained, with particular care taken not to look directly into the path of the laser. The hardware was activated and deactivated in sequence, including reconnecting the computer's network cable for the duration of the experiment, and closing the experiment box to protect the apparatus from dust when not in use[1].

3 Data Analysis

3.1 Uncorrected Adaptive Optics System

In order to characterise the static wavefront error of the system, the Peak-to-Valley (P2V) and Root Mean Square (RMS) were recorded as a function of time, and the corresponding science image recorded. This was initially done with no correction to establish a baseline against which the rest of the experiment could be compared with.

The measured P2V value was determined to be 12.81 $\pm$ 0.02 μm with an RMS of 3.269 $\pm$ 0.005 μm. These values are also given in Table 1, along with the Strehl ratio, S, and the Full-Width-Half-Maximum (FWHM) in x and y coordinate directions.

The Strehl ratio S corresponds to the ratio of the intensity at the centre of a PSF to the height of a perfect Airy function's PSF[1]. A larger value of S, within the limit $0 \leq S \leq 1$, thus indicates an improvement to an optical system's image quality[1]. In this experiment, the Strehl ratio was calculated using the equation:

$$S = \exp\left[-\left(\frac{2\pi\sigma}{\lambda}\right)^2\right] \tag{2}$$

where σ is the RMS of the wavefront error, and $\lambda = 635nm$ is the wavelength of the laser[3]. From error propagation, the uncertainty on the Strehl ratio ΔS was then given by:

$$\Delta S = 2\sigma\left(\frac{2\pi\sigma}{\lambda}\right)^2 S\Delta\sigma \tag{3}$$

where $\Delta\sigma$ was the uncertainty on the RMS of the WFE, given by its standard deviation. For the uncorrected science-camera image, shown in Fig. 3, the Strehl ratio was determined to be 0.0 ± 0.0, indicating that the image quality of the system was very poor, but also highlighting the constraints on the precision of the software.

State of wavefront	RMS (μm)	P2V (μm)	Strehl Ratio S	FWHM [x] (pix)	FWHM [y] (pix)
Uncorrected	3.269 ± 0.005	12.81 ± 0.02	0.0 ± 0.0	25.0 ± 1.0	35.0 ± 1.0
Manual Correction	0.34 ± 0.01	0.076 ± 0.003	0.5683 ± 0.0002	30.0 ± 1.0	34.0 ± 1.0
Automatic Correction	0.011 ± 0.002	0.073 ± 0.008	0.9890603 ± 0.0000004	35.0 ± 1.0	50.0 ± 1.0

Table 1: Measured wavefront error parameters for a spot produced by an incident 635 nm laser, when the wavefront was: uncorrected; manually-corrected; and automatically-corrected using the adaptive optics (AO) system's training loop. The determined parameters included the RMS and P2V WFE, the Strehl ratio S, and the Full-Width-Half-Maximum in x and y. The uncertainties on the RMS and P2V were given by their standard deviation, and the Strehl ratio and uncertainty were calculated using Eqn. 2 and 3 [3].

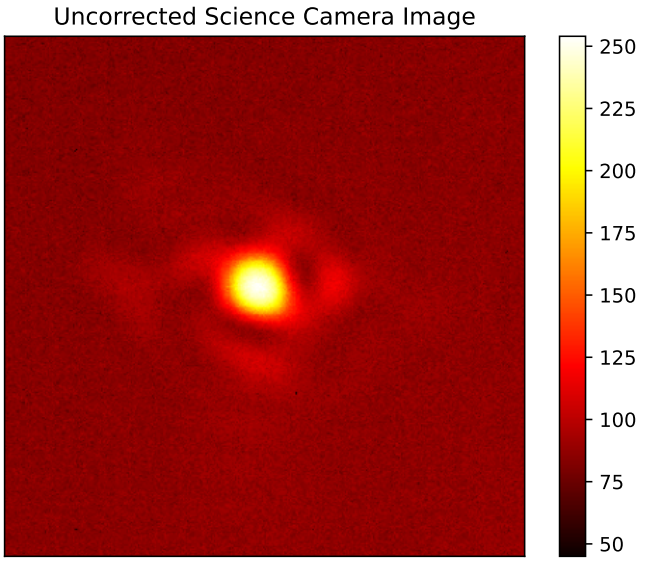


Figure 3: The science-camera image from the uncorrected AO system, centred on the spot produced by the 635 nm laser.

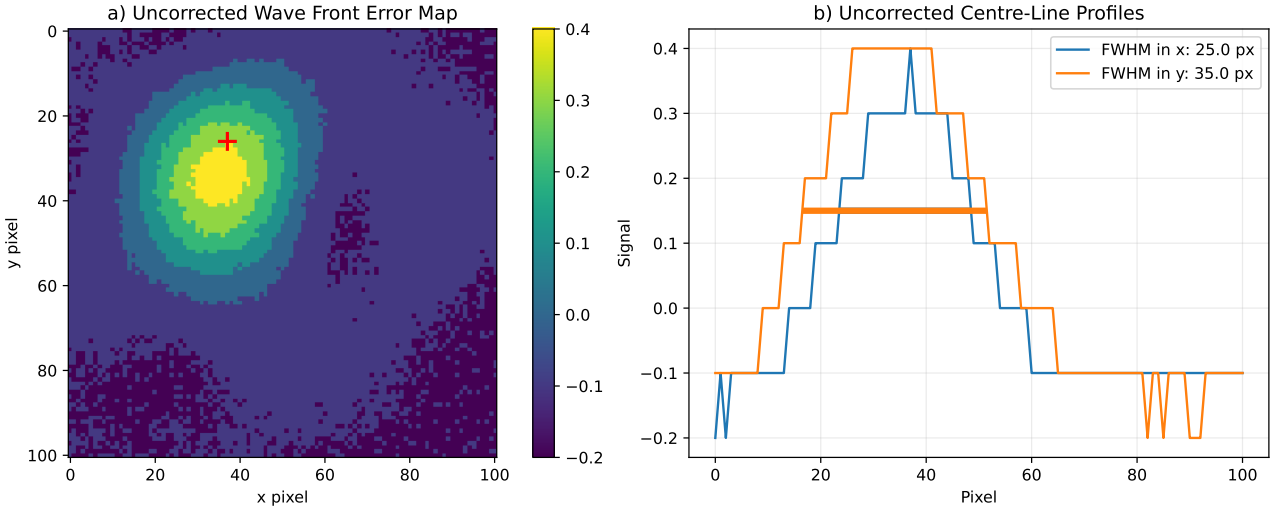
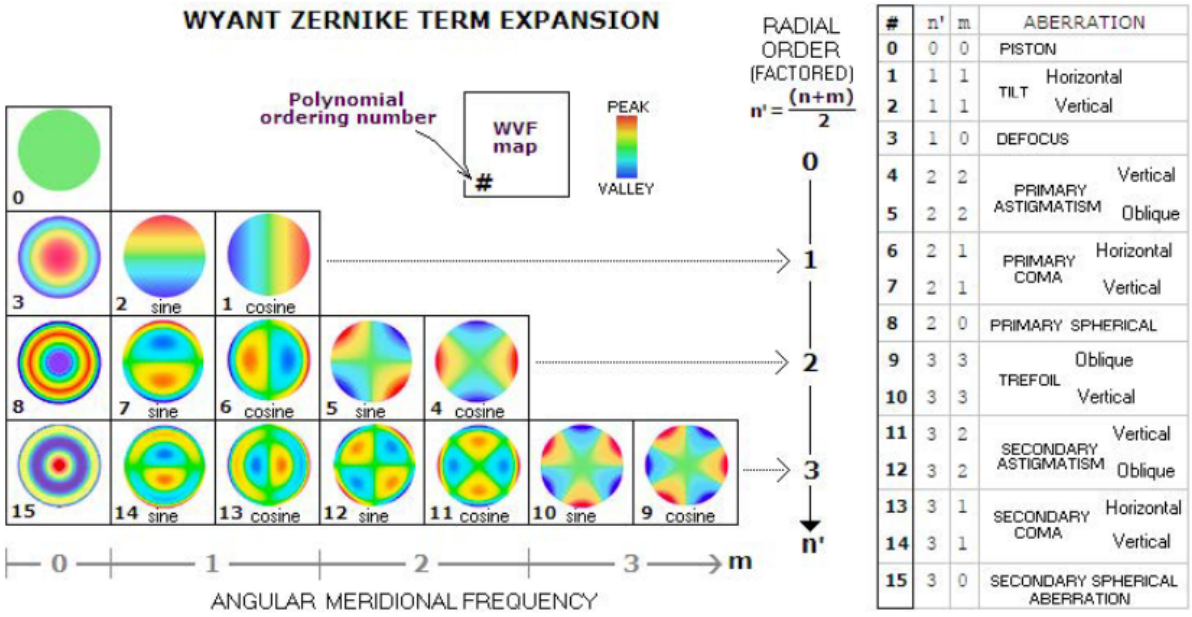


Figure 4: a) The WFE map produced by the uncorrected AO system, centred on the spot produced by the 635 nm laser. b) The corresponding centre-line profiles along the x- & y-axis of this image. From these profiles, the FWHM was determined to be 25.0 ± 0.1 pixels in x, and 35.0 ± 1.0 pixels in y.

There is a slightly elongated maximum in the science camera, which can also be seen via the WFE map in Fig. 4.a. Using centre-line profiles, as in Fig. 4.b., the FWHM was determined to be 25 ± 1 pixels in the x-direction and 35 ± 1 pixels in the y-direction. The spot was 40% broader in the vertical direction, indicating the need for correction as the spatial widths along the two axes were asymmetrical, unlike as expected in an ideal circularly symmetric profile.

3.2 Manually-Corrected Adaptive Optics System

The ability of the adaptive optics system to introduce corrections to the static wavefront error was then investigated by manually adjusting the Zernike coefficients in the control programme. The Zernike coefficients corresponding to the horizontal tilt (Z1), the vertical tilt (Z2), and the defocus (Z3), as per the scheme in Fig. 5, were altered from 0.0 μm to be 2.7 μm, -2.29 μm, and 0.05 μm respectively[1].



#	n'	m	ABERRATION
0	0	0	PISTON
1	1	1	TILT Horizontal
2	1	1	TILT Vertical
3	1	0	DEFOCUS
4	2	2	PRIMARY ASTIGMATISM Vertical
5	2	2	PRIMARY ASTIGMATISM Oblique
6	2	1	PRIMARY COMA Horizontal
7	2	1	PRIMARY COMA Vertical
8	2	0	PRIMARY SPHERICAL
9	3	3	TREFOIL Oblique
10	3	3	TREFOIL Vertical
11	3	2	SECONDARY ASTIGMATISM Vertical
12	3	2	SECONDARY ASTIGMATISM Oblique
13	3	1	SECONDARY COMA Horizontal
14	3	1	SECONDARY COMA Vertical
15	3	0	SECONDARY SPHERICAL ABERRATION

Figure 5: Graphical display of Zernike term expression [1]. Each image shows the result of one of the first 15 Zernike polynomials.

As shown in Table 1, the resulting science image was determined to have a RMS WFE of 0.34 ± 0.01 μm, and a Strehl ratio of 0.5683 ± 0.0002. Compared to the previous Strehl ratio of effectively 0 for the uncorrected system, this indicates that merely manually adjusting the optics system has greatly improved the image quality. The Strehl ratio measures the peak of the central intensity in relation to a perfect Airy pattern and the FWHM measures its apparent width; as such these two do not need to improve at the same rate for the image improvement to be a success. The science image and its centre-line profiles can be seen in Fig. 6, with the FWHM now being 30.0 ± 1.0 pixels in x and 34.0 ± 1.0 pixels in y. It should be noted that, while the image quality has been improved, there is now a noted asymmetry in the science image, which can be seen in Fig. 6.a.

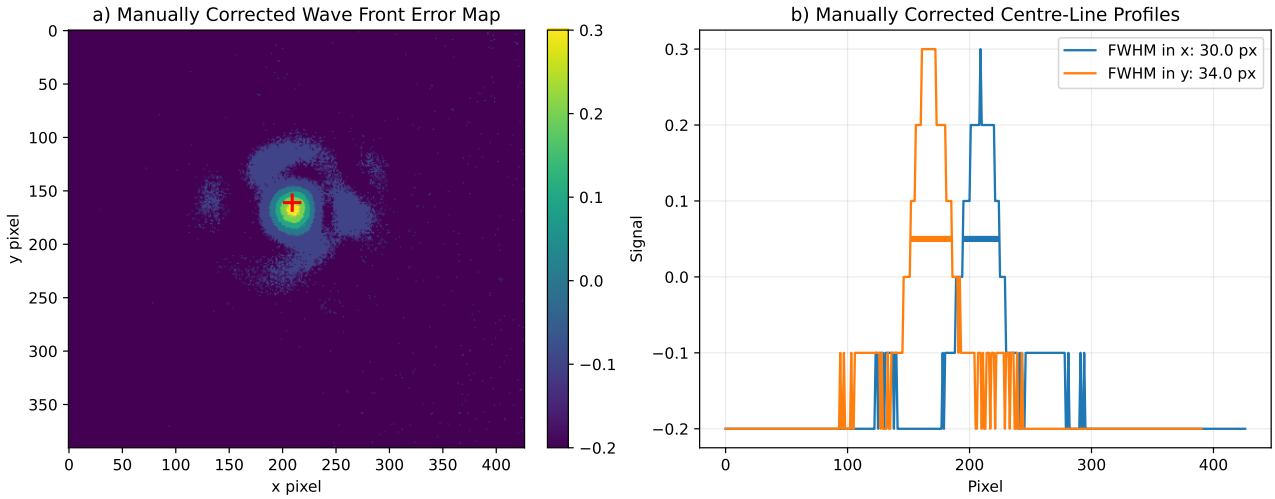


Figure 6: a) The WFE map produced by the manually-corrected AO system, centred on the spot produced by the 635 nm laser. b) The corresponding centre-line profiles along the x- & y-axis of this image. From these profiles, the FWHM was determined to be 30.0 ± 1.0 pixels in x, and 34.0 ± 1.0 pixels in y.

3.3 System Stability for the Manually-Corrected Adaptive Optics System

After manual WFE corrections, the system stability was then tested using over 13 snapshots taken 10 ± 1 seconds apart. Zernike coefficients Z1, Z2, Z3, and Z5 showed fluctuations whilst the higher order corrections showed no variation at the recorded precision during the sample time. Figure 7 shows that any varying coefficient has a peak-to-peak range of $|0.01|$ μm. The absence of a clear, continuous drift of the coefficients indicates that the correction was stable. Intermittent changes could have arisen due to the joint interference of the overhead light and movement of the group within the room. This could have created slight perturbations in the air that resulted in the short-lived changes in coefficient values. However, this apparent consistency is limited by the precision of the system. Additionally, this temporal stability does not imply that the remaining WFE was small as non-zero aberrations might have stayed constant throughout the measurement.

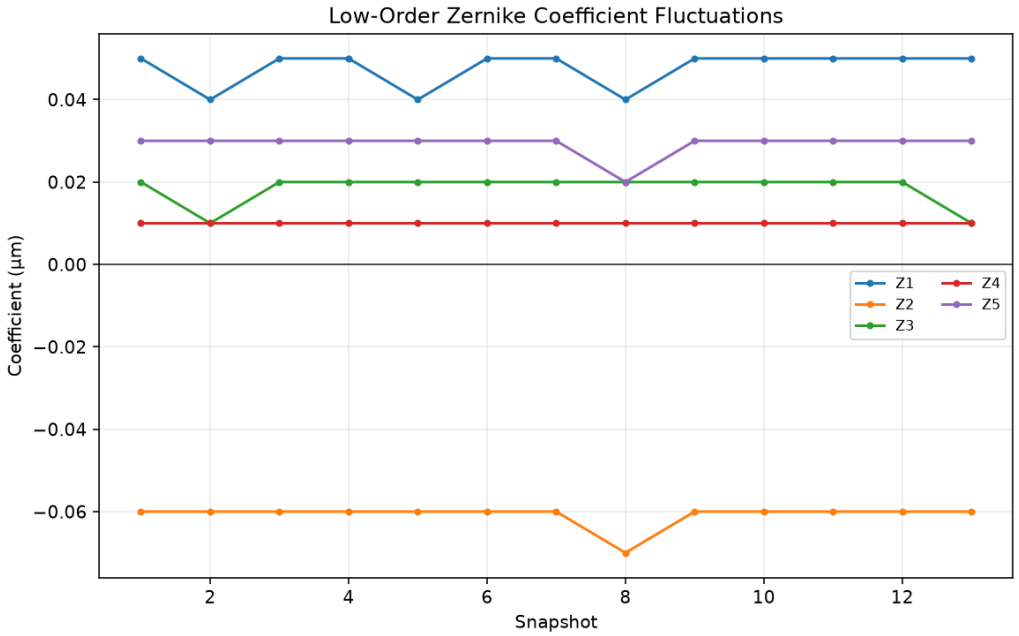


Figure 7: The variation of the Zernike coefficients over the course of the 13 snapshots each taken approximately 10 seconds apart. Coefficients for the polynomials past 5 are omitted as they remained constant as far as instrumental precision is concerned.

3.4 Automatically-corrected Adaptive Optics System

The static WFE was then automatically-corrected by setting up and running the training loop in the control programme. This loop produced the reaction matrix required to automatically adjust the deformable mirror based on the detections of the wavefront sensor[1]. Upon completion, the loop of wavefront correction was closed in the control programme, and the corrected static WFE and corresponding science image were documented.

The determined WFE parameters such as the RMS, P2V, Strehl ratio, and FWHM are again given in Table 1. For an image produced by the adaptive optics system automatically correcting itself, the Strehl ratio S was 0.9890603 ± 0.0000004, which indicates a significant improvement to the image quality compared to when the system was manually corrected ($S = 0.5683 \pm 0.0002$) and when it was uncorrected ($S = 0.0 \pm 0.0$). The Strehl ratio for the automatically-corrected system was nearly of order unity, with $S = 1$ corresponding to a system with perfect image quality[1].

The improvement in image quality can be seen in Fig. 8.a., where the science-camera image exhibits a close approximation of an Airy pattern, with a noted elongation along the y-axis[1]. This elongation can also be observed from the FWHM values, which were determined from the centre-line profiles in Fig. 8.b. to be 35.0 ± 1.0 pixels in x and 50.0 ± 1.0 pixels in y, and indicates potential astigmatism within the optical system[1].

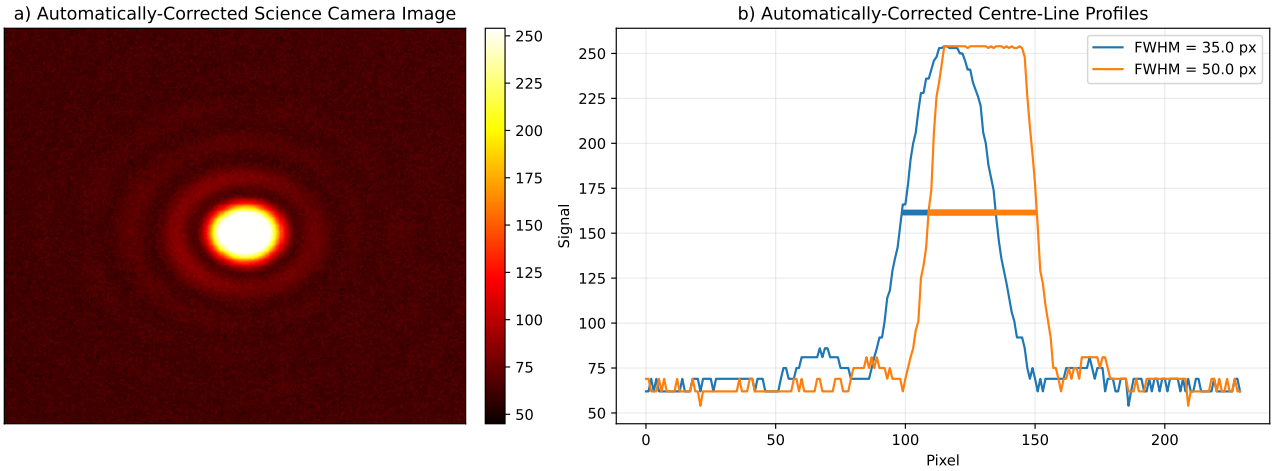


Figure 8: a) The science-camera image from the automatically-corrected AO system, centred on the spot produced by the 635 nm laser. b) The corresponding centre-line profiles along the x- & y-axis of this image. From these profiles, the FWHM was determined to be 35.0 ± 1.0 pixels in x, and 50.0 ± 1.0 pixels in y.

The system stability was again tested by recording the Zernike coefficients every 10 ± 1 seconds. Within the system's precision of 0.01 μm, all of the Zernike coefficients, from order Z0 to Z20, were recorded as 0.00 μm. While the displayed science-camera image and WFE map were observed to be very stable after the adaptive optics system was set to automatically correct, the consistent zero values observed for the Zernike coefficients were likely due to the limitations of the system's precision. Further instrument precision would allow better assessment of the system's stability, and enable us to better correct and investigate observed aberrations, such as the elongation along the y-axis seen in Fig. 8.a.

3.5 Science-Image Quality Under Applied Aberrations

In order to better understand the effects of individual and combined Zernike transformations on the quality of the science image, the automatic-correction training loop was utilised to apply specific input Zernike transformations. Because the resulting science images strongly deviate from the Gaussian distribution, using FWHM in order to gauge the quality of the resulting image gives unsatisfactory results, especially given the spatially separated peaks that occur in many of these images. The method used here in order to quantify the level of distortion in the resulting images was to determine the radius from the peak intensity at which a certain proportion of the total intensity of the image is contained. This gives better values when the image is more constrained and more circular, so the closer an image is to the expected Airy profile of an undistorted image, the tighter the radius of the circle constraining a certain portion of the total intensity will be.

In Fig. 9 we see the results of this method. The top left image shows the image with the vertical trefoil (Z10) at 0.1 μm: this resulted in FWHMs of 58.0 ± 1.0 and 45.0 ± 1.0 pixels on the x- and y-axis respectively. The circle containing 90% of the intensity in the image had a radius of 60 ± 1.0 pixels, which is broadly consistent with the FWHM in indicating that the image is distorted, but gives a value that indicates the dispersion of the data more clearly. The

top right image utilised a value of 0.1 defocus (Z3 = 0.1 μm), which results in an image that looks fine with visual inspection. However, the value of 39.0 ± 1.0 pixels for the radius of the circle shows a wide spread for the image.

The bottom left image in Fig. 9 shows the result of 0.1 vertical primary coma (Z7 = 0.1 μm) and 0.1 vertical primary astigmatism (Z5 = 0.1 μm). This image similarly results in a radius of 39 ± 1.0 pixels, indicating a similar over concentration of intensity when compared to the defocus image. This is due to a significant portion of the intensity being focused at peaks trailing away from the centre of the image. The bottom right image shows the result of 0.1 primary spherical (Z8 = 0.1 μm) and 0.1 oblique trefoil distortions (Z9 = 0.1 μm). This results in a significantly larger circle with a radius of 72 ± 1.0 pixels being required to encompass 90% of the total intensity as there are peaks with significant intensity leading away from the centre of the image where the main peak is located.

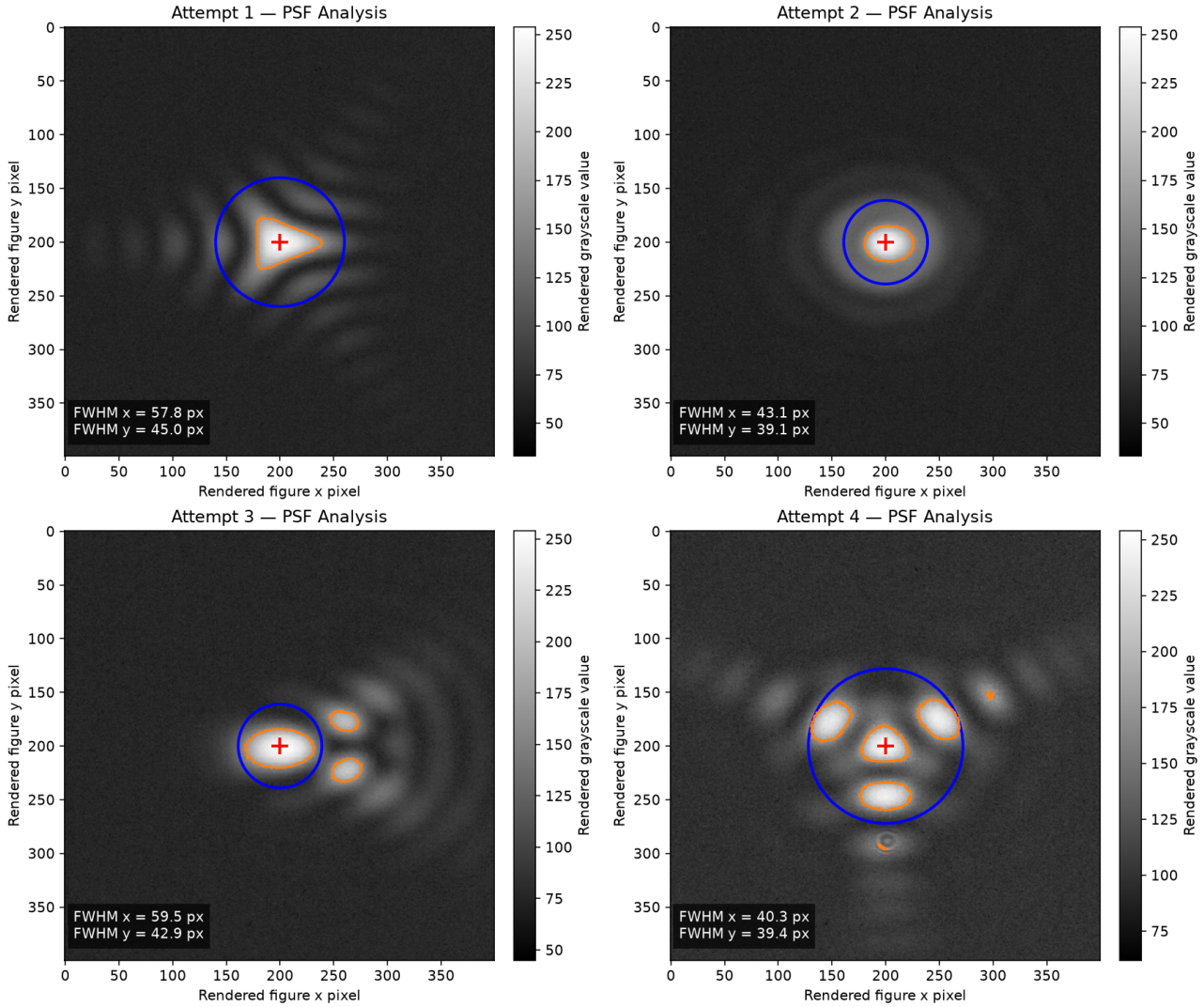


Figure 9: The resulting science images after several distinct Zernike polynomials were applied to the wavefront. The blue circles encompass an area within which 90% of the total intensity of the image is located. The orange line traces the location where the data is 50% of the maximum intensity in each image.

A summary of the parameters such as WFE RMS, Strehl ratio S, and FWHM in x and y, that were determined for the attempts discussed above are also recorded in Table 2. While using a bounding radius to determine the size of the images works well, it gives limited information about the shapes caused by the aberrations. Despite the visible aberrations in the images, the error values and technical quality remained similar to those of images resulting from corrected wavefronts. It should be noted though that there is a pattern of larger RMS values and lower Strehl ratios as more aberrations are added, as can be seen in Table 2.

Attempt	Altered Zernike Coefficient (μm)	RMS (μm)	P2V (μm)	Strehl ratio S	FWHM [x] (pix)	FWHM [y] (pix)	Profile-based radius (rendered pix)
1	Z10 = 0.1	0.100 ± 0.002	0.552 ± 0.001	0.3757 ± 0.0002	58.0 ± 1.0	45.0 ± 1.0	60 ± 1.0
2	Z3 = 0.1	0.100 ± 0.001	0.375 ± 0.003	0.37566 ± 0.00007	43.0 ± 1.0	39.0 ± 1.0	39 ± 1.0
3	Z5 = 0.1 Z7 = 0.1	0.139 ± 0.004	0.879 ± 0.008	0.1508 ± 0.0003	60.0 ± 1.0	43.0 ± 1.0	39 ± 1.0
4	Z8 = 0.1 Z9 = 0.1	0.143 ± 0.003	0.701 ± 0.006	0.1351 ± 0.0002	40.0 ± 1.0	39.0 ± 1.0	72 ± 1.0

Table 2: Provisional FWHM estimates, profile-based radii, and estimated Strehl ratios for the four retained attempts, after several distinct Zernike polynomials were applied to the wavefront. For each attempt, the Zernike coefficients which were altered from 0.0 μm to 0.1 μm are identified per the scheme outlined in Fig. 5.

3.6 Wavefront Correction with a Turbulence Simulator

In order to approximate real-life changes in the atmosphere, a turbulence simulator was placed in front of the free path region behind the first collimator. The WFE was measured both before and after the automatic correction. Table 3 shows the drastic difference the adaptive optics system makes in correcting for the added turbulence. Both P2V and RMS reduced by $\sim$ 94%. Importantly, however, the correction reduced the WFE but it did not eliminate it as both metrics still have residuals, 0.835μm and 0.213μm respectively. The standard deviation of both show a reduction of $\sim$ 30%, highlighting that the reduction in mean error is significantly larger than the reduction in its fluctuations.

Figure 10 depicts the science camera array for the corrected image and the corresponding horizontal and vertical intensity profiles. The distribution has an elongated centre with smaller bright spots surrounding it, rather than a single, compact circular spot. The estimated FWHM is 30 pixels in the x direction and 85 pixels in the y direction. However, the exported data was strongly quantised and hence should be taken as approximate descriptors of the distribution rather than precise measures of optical resolution. Since an equivalent uncorrected science-camera image was not successfully recorded, the improvement in PSF width cannot be quantified. Therefore, the result shows the improved WFE, but does not establish the magnitude of the improvement in the science-camera image quality.

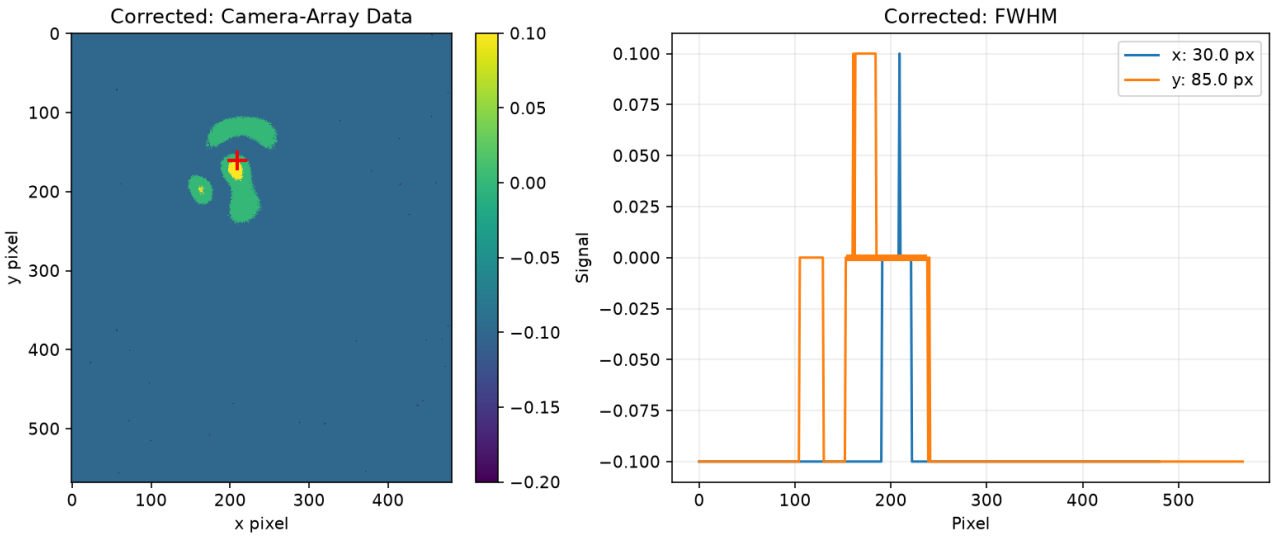


Figure 10: a) The corrected science-camera image with the turbulence simulator in place. The red lines indicate the peak intensity. b) The corresponding centre profiles along the x- & y-axis of this image. From these profiles, the FWHM was determined to be 30.0 ± 1.0 pixels in x, and 85.0 ± 1.0 pixels in y.

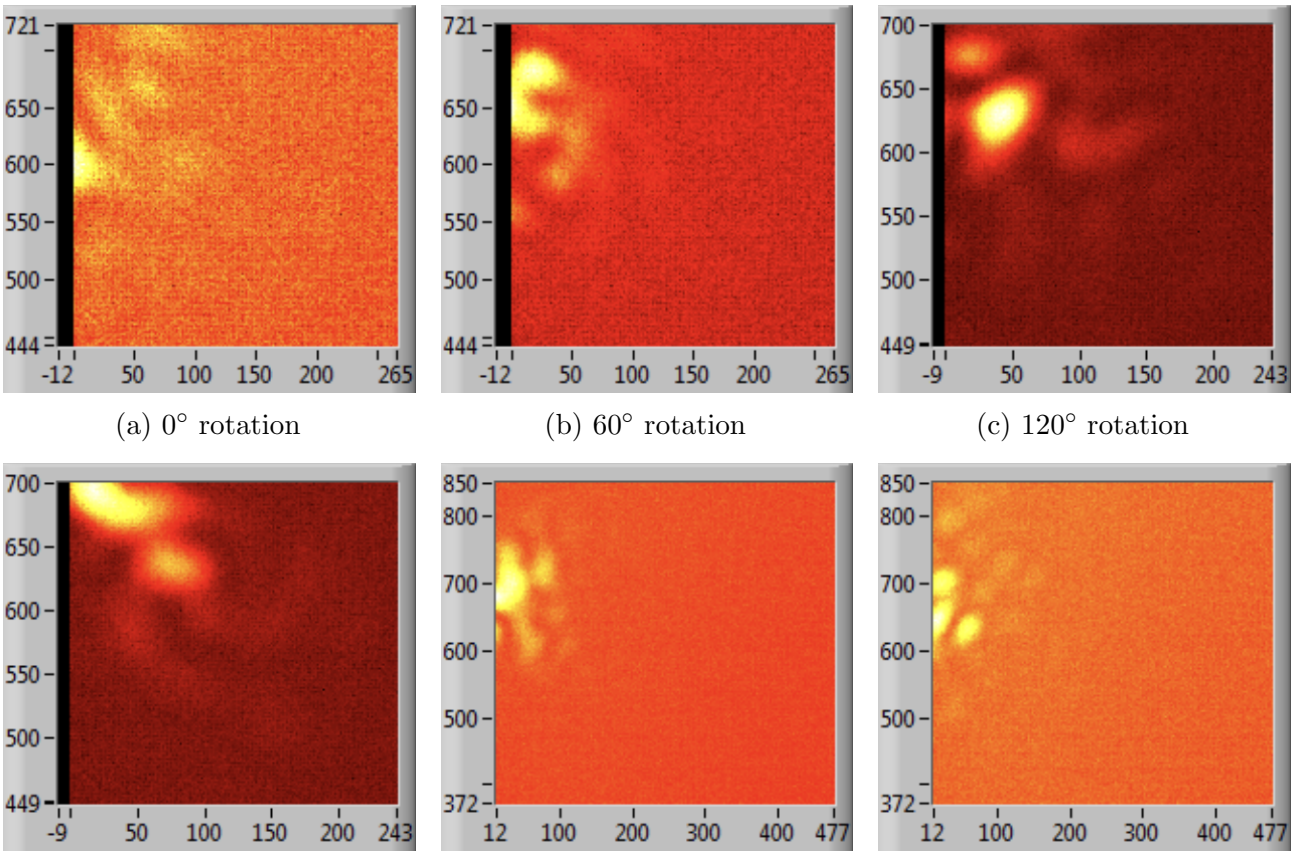
State	P2V WFE (μm)	P2V SD (μm)	RMS WFE (μm)	RMS SD (μm)
Uncorrected	14.5541	0.0975	3.75018	0.03338
Corrected	0.8348	0.0693	0.21306	0.02347

Table 3: Peak-to-valley (P2V) and RMS wavefront errors and their standard deviations (SD) with the turbulence simulator in place, before and after automatic-correction by the adaptive optics system.

To investigate performance under time-varying aberrations, the turbulence disc was rotated slowly while the wavefront error history and science-camera image were recorded. Table 4 depicts how the WFE changed as the simulator was rotated $\sim 60°$ of a full rotation. The WFE initially decreased, reaching its minimum, P2V 14.8μm and RMS 3.8μm, at 120° of full rotation, after which both metrics generally increased again. Figure 11 shows the corresponding science-camera images, which display varying, irregular intensity distributions, with parts of the spot outside the recorded frame.

Rotation	P2V WFE (μm)	RMS WFE (μm)
0°	18.098	4.639
60°	16.071	3.936
120°	14.818	3.795
180°	16.583	4.268
240°	16.167	4.197
300°	17.675	4.623

Table 4: Instantaneous P2V and RMS wavefront-error readouts recorded during rotation of the turbulence simulator.



(a) 0° rotation (b) 60° rotation (c) 120° rotation

(d) 180° rotation (e) 240° rotation (f) 300° rotation

Figure 11: Science-camera PSF screenshots recorded during rotation of the turbulence simulator. Angles are approximate and measured relative to the starting position. The displayed coordinate ranges differ between panels, and parts of the spot lie outside the recorded field in some images.

4 Conclusion

This investigation looked at wavefront measurements and corrections using an adaptive optics system and a 635nm laser. In its uncorrected state the wavefront exhibited a large error with an RMS WFE of 3.269 $\pm$ 0.005 μm and P2V WFE of 12.81 $\pm$ 0.02 μm. The corresponding science-camera image was distorted, with FWHM values of 25.0 $\pm$ 1.0 pixels in the x direction and 35.0 $\pm$ 1.0 pixels in the y direction.

Manual adjustment of the Zernike coefficients reduced the RMS and P2V WFE to 0.34 ± 0.01 μm and 0.076 ± 0.003 μm, respectively. Additionally, the Strehl ratio increased to 0.5683 ± 0.0002, a substantial increase from the initial effectively 0 value in the uncorrected system. The FWHM stayed broadly similar with a slight increase in the x direction to 30.0 ± 1.0 pixels and staying roughly the same in the y direction with 34.0 ± 1.0 pixels. This highlights how the Strehl ratio and FWHM describe different aspects of image quality and do not need to improve at the same rate.

The manually-corrected system was found to be stable over 13 measurements separated by approximately 10 seconds each at the software precision. The Zernike coefficients remained within peak-to-peak range of $|0.01|$ μm, with no clear continuous drift. This suggests a stable correction, however does not rule out non-zero stable aberrations.

Automatic correction utilised the system's training loop and resulted in the largest improvement. WFE decreased to an RMS of 0.011 ± 0.002 μm and a P2V of 0.073 ± 0.008 μm, yielding a Strehl ratio of 0.9890603 ± 0.0000004. This confirms that the adaptive optics system corrected for the WFE effectively, with the result being close to an Airy pattern. However, the resulting image retained an elongation in the y-axis, as the FWHM in the x and y direction were 35.0 ± 1.0 pixels and 50.0 ± 1.0 pixels, respectively. This indicates potential astigmatism within the optical system[1].

When the stability of the automatically-corrected system was investigated, all of the Zernike coefficients from order Z0 to Z20 were recorded as being 0.00 μm, indicating perfect stability. The system's limited precision of 0.01 μm does however mean that, while the displayed science-camera image and WFE map were observed to be very stable, the consistent zero values observed for the Zernike coefficients are likely due to system limitations.

The quality of the science-camera images were then investigated in the case of specific aberrations, created via altering the value of specific Zernike coefficients from 0.0 μm to 0.1 μm, and then utilising the automatic-correction training loop to generate corresponding wavefront errors. This was done to create aberrations such as: vertical trefoil ($Z10$); defocus ($Z3$); combined vertical primary coma ($Z7$) and vertical primary astigmatism ($Z5$); and combined primary spherical ($Z8$) and oblique trefoil ($Z9$)[1]. When these aberrations were deliberately introduced, the resulting distributions became strongly non-Gaussian, and in some cases had multiple peaks. For these cases, the FWHM was deemed an inadequate image quality indicator, and instead a radius enclosing 90% of the total image intensity provided a more informative measure of the spread. Despite the visible aberrations in the images, the error values and technical quality remain similar to those of images resulting from corrected wavefronts. It was also observed that, as more aberrations were added, the WFE RMS values increased, with the Strehl ratios similarly decreasing as the science-camera images deviated further from that of an ideal Airy function[1].

Finally, a turbulence simulator was introduced to the adaptive optics system, which increased the WFE substantially. Automatic corrections reduced both RMS and P2V WFE by $\sim 94\%$. However, WFEs of 0.213 μm in RMS and 0.835 μm in P2V remained, showing that the AO system reduced but did not eliminate the aberrations. As an equivalent science-camera image for the uncorrected system was not recorded, the level of improvement in PSF could not be quantified.

Overall, the investigation demonstrated that an adaptive optics system can reduce both static and turbulence-induced wavefront errors. The experiment was limited by residual aberrations, the precision of the measurements, and the absence of a directly comparable uncorrected science-camera image for the turbulence image.

References

[1] Kerstin Seiffarth et al. *Adaptive Optics (AO)*. 2021. URL: https://www.usm.lmu.de/Lehre/Lehrveranstaltungen/Praktikum/materials/ao/AO_0.02_00_017_211020.pdf. (frank@grupp-astro.de) V0.02. (accessed: 15/10/2026).

[2] Mohamed E. Shaheen et al. "Determination of the refractive index of air and its variation with temperature and pressure using a Mach-Zehnder interferometer". In: *Journal of Optics* 53.3 (Nov. 2023), pp. 2219–2228. DOI: 10.1007/s12596-023-01486-2.

[3] Robert K. Tyson. "Introduction to adaptive optics". In: Bellingham, WA : SPIE Press, 2000. Chap. Chapter 2: "Adaptive optics systems — *optics* is our middle name", pp. 19–24. DOI: 10.1117/3.358220.